

Absolutely — **acceleration** is a good one because it spans **physics, vehicle/body movement, sport performance, rate of change/growth, business/economic acceleration, and the broader accelerate / accelerator family**.

I'll keep the same depth and apply the same Golden Manual rule: broad search across literal, technical, figurative and family uses, but merge uses that share the same core concept.

Word family: **accelerate / accelerates / accelerated / accelerating / acceleration / accelerator / accelerative**

Base verb: accelerate

Related noun: acceleration

Person/device noun: accelerator

Related adjective: accelerated

Less common technical adjective: accelerative

Useful compounds / extensions: acceleration mechanics, acceleration phase, rate of acceleration, angular acceleration, linear acceleration, gravitational acceleration, accelerated growth, accelerated learning, accelerated programme, hardware accelerator

Part of speech: verb / noun / adjective-related forms

Your sentence

> "...a sprinter returning from a hamstring injury may need a progressive programme built around mobility and **acceleration mechanics**..."

Meaning in this passage

Here, **acceleration** means the phase and process in which a sprinter **increases running speed**, especially after beginning from rest or a relatively low speed.

In sprinting, acceleration involves progressively increasing velocity through coordinated:

- * force production
- * body position
- * stride length
- * stride frequency
- * foot placement
- * arm action
- * posture

繁體中文：

運動員由較低速度逐步提升至更高速度的過程；在短跑中尤其指起跑後逐漸加快速度的階段

中文翻譯：

> 「.....從腿後肌受傷中復出的短跑運動員，可能需要一套以活動能力和**加速動作技術**為核心

的循序漸進訓練計劃.....」

****acceleration mechanics = 短跑運動員在提升速度階段所採用的姿勢、步態、發力方向和身體協調方式****

1. acceleration — increase in speed — 加速；速度增加

The broad everyday physical meaning.

Practice 1

> “The car’s sudden ****acceleration**** pushed me back into my seat.”

「汽車突然****加速****，令我的身體向後壓在座位上。」

Practice 2

> “The cyclist showed impressive ****acceleration**** on the final hill.”

「單車手在最後一段斜坡展現出很強的****加速能力****。」

****acceleration = 某人 / 物的移動速度變得更快的過程****

This is the everyday interpretation.

2. acceleration — physics: rate of change of velocity — 加速度

The precise scientific meaning is broader than simply “getting faster.”

In physics, ****acceleration**** means the ****rate at which velocity changes over time****.

Because velocity includes both speed and direction, acceleration can occur when something:

- * speeds up;
- * slows down;
- * changes direction.

Practice 3

> “The object has a constant ****acceleration****.”

「這個物體具有恆定的****加速度****。」

Practice 4

> “Acceleration is measured in metres per second squared.”

「****加速度****以米每二次方秒量度。」

****acceleration — physics = 物體速度向量隨時間改變的速率，包括速度大小或方向的改變****

Important learner point

Everyday:

> acceleration $\approx$ getting faster

Physics:

> acceleration = any change in velocity

So braking can involve acceleration mathematically, even though everyday English would normally call it ****deceleration****.

3. rate of acceleration — 加速率；加速度程度

Practice 5

> “The vehicle’s ****rate of acceleration**** increased.”

「車輛的****加速程度 / 加速度****提高了。」

Practice 6

> “Coaches measured the runner’s rate of acceleration from the start.”

「教練測量了跑手起步後的****加速率****。」

In ordinary sporting language this often means:

> how quickly someone gains speed.

4. acceleration — sporting ability to gain speed — 加速能力

This is especially important in sprinting and field sports.

Practice 7

> “Good **acceleration** is essential for a sprinter.”

「良好的**加速能力**對短跑運動員非常重要。」

Practice 8

> “The training improved her acceleration over the first thirty metres.”

「訓練改善了她首三十米的**加速能力**。」

acceleration — sport = 運動員在短時間內由低速提升至高速的能力

This differs subtly from:

top speed = 最高速度

An athlete can have excellent acceleration without having the highest maximum speed.

5. acceleration phase — 加速階段

A very useful sprinting compound.

Practice 9

> “The athlete maintained a low body angle during the early **acceleration phase**.”

「運動員在早段**加速階段**保持較低的身體角度。」

Practice 10

> “The acceleration phase occurs before maximum velocity is reached.”

「**加速階段**發生在運動員達到最高速度之前。」

acceleration phase = 由起步 / 較慢速度逐步提高速度，直到接近最高速度的運動階段

6. acceleration mechanics — 加速動作機制；加速技術

Your exact expression.

Practice 11 — Passage sense

> “The programme focused on mobility and **acceleration mechanics**.”

「訓練計劃著重活動能力和**加速動作技術**。」

Practice 12

> “Poor acceleration mechanics may reduce force application.”

「不良的**加速動作機制**可能降低發力效率。」

Practice 13

> “The coach rebuilt the athlete’s acceleration mechanics after injury.”

「運動員受傷後，教練重新調整了他的**加速技術動作**。」

acceleration mechanics = 在提高跑速期間，身體姿勢、落腳、步態、發力及四肢協調所形成的技術模式

7. linear acceleration — 線加速度

In physics and engineering:

Practice 14

> “The sensor measures **linear acceleration**.”

「感應器測量**線加速度**。」

Practice 15

> “Linear acceleration changes an object’s velocity along a path.”

「**線加速度**改變物體沿路徑運動的速度。」

linear acceleration = 物體沿直線 / 平移方向的速度變化率

8. angular acceleration — 角加速度

The rate at which rotational velocity changes.

Practice 16

> “The wheel experienced increasing **angular acceleration**.”

「車輪的**角加速度**持續增加。」

Practice 17

> “Angular acceleration describes changes in rotational speed.”

「**角加速度**描述旋轉速度隨時間的變化。」

angular acceleration = 物體旋轉角速度隨時間改變的速率

9. gravitational acceleration — 重力加速度

Practice 18

> “Near Earth’s surface, gravitational **acceleration** is approximately constant for many basic calculations.”

「在基本計算中，地球表面附近的**重力加速度**通常視為近似恆定。」

Practice 19

> “The experiment measured gravitational acceleration.”

「實驗測量了**重力加速度**。」

gravitational acceleration = 物體受重力作用而產生的加速度

10. acceleration — increase in rate/intensity of a process — 加速；加快

Now the meaning becomes figurative.

Acceleration can describe something happening or developing increasingly quickly.

Practice 20

> “There has been an **acceleration** in technological change.”

「科技變革出現了明顯的**加速**。」

Practice 21

> “The company reported an acceleration in demand.”

「公司報告需求增長正在**加快**。」

acceleration = 某個發展、變化或活動進行速度變快的現象

This extends the physical concept:

> movement becomes faster

> → process becomes faster.

11. acceleration of growth — 增長加速

Practice 22

> “The economy showed an **acceleration of growth**.”

「經濟顯示出**增長加速**。」

Practice 23

> “The second half of the year saw an acceleration in sales growth.”

「下半年銷售增長出現**加快**。」

acceleration of growth = 增長本身的速度進一步提高

Important distinction

This is not simply:

> growth

It means:

> growth is becoming faster.

12. acceleration — economic/business momentum increasing — 增長加快；發展提速

Common in economics and business reports.

Practice 24

> “Analysts expect an **acceleration** in economic activity.”

「分析員預期經濟活動會**加快**。」

Practice 25

> “Investment helped drive the acceleration in expansion.”

「投資推動了擴張速度的**加快**。」

acceleration — business/economics = 經濟活動、需求、銷售、投資或增長速度提高

13. acceleration — faster progression through a programme/process — 加速進程；提前進展

In education, development, training or institutional contexts:

Practice 26

> “The programme allows **acceleration** for students who progress quickly.”

「這個課程容許進度較快的學生**加速學習 / 提前進階**。」

Practice 27

> “Acceleration may allow gifted students to complete material earlier.”

「**加速學習安排**可以讓資優學生較早完成課程內容。」

acceleration = 讓某人比正常進度更快通過一個學習 / 發展過程

14. accelerate — increase speed — 加速

The core verb.

Practice 28

> “The runner **accelerated** out of the bend.”

「跑手出彎後開始**加速**。」

Practice 29

> “The car accelerated rapidly.”

「汽車迅速**加速**。」

accelerate = 使自己 / 某物的速度提高，或本身速度變快

15. accelerate + object — make something move faster — 使加速

The verb can be transitive.

Practice 30

> “The driver **accelerated** the vehicle.”

「司機令車輛**加速**。」

Practice 31

> “The force accelerates the object.”

「這股力令物體產生**加速度 / 加速**。」

accelerate something = 令某物的速度 / 速度向量改變得更快

16. accelerate a process/change — 加快；促進

A major figurative verb use.

Practice 32

> “Technology has **accelerated** communication.”

「科技**加快了**溝通速度。」

Practice 33

> “The injury accelerated his decision to retire.”

「這次受傷**加速 / 促使他更早作出**退役決定。」

accelerate something = 令某項變化、過程或發展比原本更快發生

17. accelerate growth/development — 加速增長 / 發展

Practice 34

> “The investment helped **accelerate growth**.”

「這項投資有助**加速增長**。」

Practice 35

> “Good support can accelerate an athlete’s development.”

「良好的支援可以**加快運動員的發展**。」

This is especially relevant to coaching.

18. accelerated — made faster than usual — 加速的；加快的

Practice 36

> “The company entered a period of **accelerated growth**.”

「公司進入了**加速增長**階段。」

Practice 37

> “The injury led to accelerated rehabilitation planning.”

「這次受傷令復康規劃進入**加速安排**。」

accelerated = 以比正常或先前更快的速度進行的

19. accelerated learning — 加速學習

Practice 38

> “The course uses an **accelerated learning** format.”

「這個課程採用**加速學習模式**。」

Practice 39

> “Accelerated learning allows students to cover material more quickly.”

「**加速學習**讓學生以較快速度完成課程內容。」

accelerated learning = 以較正常課程更快的進度完成學習內容的方式

20. accelerated programme / accelerated course — 加速課程；密集課程

Practice 40

> “She completed an **accelerated programme**.”

「她完成了一個**加速 / 密集課程**。」

Practice 41

> “The accelerated course covers the same material in less time.”

「這個**加速課程**以較短時間完成相同教材。」

21. accelerator — vehicle control used to increase speed — 油門；加速踏板

A major concrete family member.

Practice 42

> “He pressed the **accelerator**.”

「他踩下**油門 / 加速踏板**。」

Practice 43

> “Take your foot off the accelerator.”

「把腳從**油門**移開。」

accelerator = 車輛中用來增加引擎輸出並通常令車輛加快的控制裝置 / 踏板

Especially British English.

American English often simply says:

> gas pedal

22. accelerator — device/system that speeds up a process — 加速器

A broader technical meaning.

Practice 44

> “The hardware **accelerator** improves processing speed.”

「硬件**加速器**提升了處理速度。」

Practice 45

> “The system uses an accelerator for graphics processing.”

「系統使用**加速器**處理圖像。」

accelerator = 專門令某個物理、電腦或技術過程進行得更快的裝置 / 系統

23. particle accelerator — 粒子加速器

A major scientific compound.

Practice 46

> “The laboratory operates a particle **accelerator**.”

「實驗室運作一台**粒子加速器**。」

Practice 47

> “Particle accelerators increase the energy of charged particles.”

「**粒子加速器**提高帶電粒子的能量。」

particle accelerator = 利用電磁場等方法使帶電粒子達到極高速度 / 能量的科學設備

24. hardware accelerator — 硬件加速器

Practice 48

> “The chip includes a machine-learning **accelerator**.”

「晶片內置一個機器學習**硬件加速器**。」

Practice 49

> “Hardware accelerators perform certain computations more efficiently.”

「**硬件加速器**能更有效率地處理某些特定運算。」

25. business accelerator / startup accelerator — 創業加速器；企業加速計劃

A major modern business compound.

A **startup accelerator** is a programme that helps young companies grow faster through:

- * mentoring
- * funding
- * networking
- * training
- * resources

Practice 50

> “The startup joined a technology **accelerator**.”

「這家初創企業加入了一個科技**創業加速計劃**。」

Practice 51

> “The accelerator provides funding and mentoring.”

「這個**創業加速器 / 加速計劃**提供資金和導師支援。」

****startup accelerator = 透過資金、導師、網絡和密集支援幫助初創企業快速發展的計劃 / 機構****

This is a lexicalized modern use.

26. acceleration programme — 加速發展計劃

Common in business and institutional contexts.

Practice 52

> “The organisation launched a leadership ****acceleration programme****.”

「機構推出了領導人才****加速發展計劃****。」

Practice 53

> “The programme accelerates promising employees into senior roles.”

「這項計劃協助有潛質的員工****加快晉升至高級職位****。」

This is usually a transparent compound rather than a new meaning.

27. accelerator — person/factor that speeds development — 加速因素；促進者

Sometimes figuratively used for something that speeds change.

Practice 54

> “Technology acted as an ****accelerator**** of change.”

「科技成為變革的****加速因素****。」

Practice 55

> “The crisis was an accelerator for digital adoption.”

「這場危機成為數碼化採用的****加速因素****。」

****accelerator = 令某種發展、改變或採用速度明顯提高的因素****

28. deceleration — 減速；負加速度

An important contrastive family member.

Practice 56

> “The athlete practised both acceleration and **deceleration**.”

「運動員同時練習**加速和減速**。」

Practice 57

> “Poor deceleration control can increase loading on the body.”

「不良的**減速控制**可能增加身體負荷。」

deceleration = 移動速度降低的過程；在運動語境中指控制身體降低速度的能力

This is especially useful in sport.

Important distinctions

acceleration vs speed

speed

= how fast something is moving at a particular moment

acceleration

= how quickly its velocity is changing

Example:

A car can travel at:

> 100 km/h

with:

> zero acceleration

if its speed and direction remain constant.

So ****high speed does not automatically mean high acceleration****.

acceleration vs velocity

velocity

= speed with direction

acceleration

= change in velocity over time

Therefore changing direction can produce acceleration even when speed remains constant.

acceleration vs top speed

Especially relevant in sprinting.

acceleration

= how quickly an athlete gains speed

top speed

= highest speed the athlete ultimately reaches

A sprinter may have:

* exceptional acceleration;

* only average top speed;

or the reverse.

acceleration vs explosiveness

acceleration

measurable increase in movement velocity

explosiveness

broader sports quality involving very rapid force/power production

Explosiveness can contribute to acceleration, but the terms are not identical.

acceleration vs power

power

ability to produce work/force rapidly

acceleration

actual rate at which velocity changes

A powerful athlete may accelerate well, but **power** is a capacity while **acceleration** describes the resulting motion/change in velocity.

acceleration vs momentum

Very important.

acceleration

change in velocity

momentum

depends on mass and velocity and describes quantity of motion

An object can have high momentum while having zero acceleration.

They are not synonyms.

acceleration vs pace

In running:

pace

how fast someone is currently covering distance

acceleration

the process of increasing that speed

> “Her pace was fast.”

= 她當時跑得快。

> “Her acceleration was excellent.”

= 她提升速度的能力很強。

accelerate vs speed up

These overlap strongly.

speed up

common everyday expression

accelerate

more formal/technical and especially common in:

- * driving

- * physics

- * sport

- * economics

- * business

> “The car sped up.”

everyday.

> “The car accelerated.”

more technical/precise.

accelerate vs improve

accelerate development

= make development happen faster

improve development/outcome

= make it better

Something can accelerate without improving quality.

Fast ≠ better.

accelerated vs rapid

accelerated

implies something has become faster than before or than normal

rapid

simply means fast

> rapid growth

= growth is fast

> accelerated growth

= growth has sped up / is occurring at an increased rate

This distinction is particularly useful in economic writing.

accelerator vs engine

engine

provides motive power

accelerator

controls/increases the vehicle's power/speed request

Pressing the accelerator affects how much power is demanded; the accelerator itself is not the engine.

accelerator vs incubator

In startups:

accelerator

typically aims to speed growth over an intensive, relatively structured programme

incubator

often supports very early-stage businesses over a potentially longer, more developmental period

There is real-world overlap, but the names emphasize different models.

Important grammar note

****Acceleration**** is usually uncountable when referring generally to the phenomenon:

> “Acceleration is important in sprinting.”

But it can be countable in technical or event-specific contexts:

> “The sensor recorded several sudden accelerations.”

Here:

****accelerations = separate instances of increasing/changing velocity****

For general learner use, however, ****acceleration**** is most often treated as uncountable.

Family-extension sweep

After the full physics, sporting, vehicle, economic, educational, computing, business, compound and second-pass audit, the useful modern family includes:

* ****acceleration — increase in speed**** = 加速

* **acceleration — physics** = 加速度
 * **acceleration — sporting ability** = 加速能力
 * **acceleration phase** = 加速階段
 * **acceleration mechanics** = 加速動作機制 / 技術
 * **linear acceleration** = 線加速度
 * **angular acceleration** = 角加速度
 * **gravitational acceleration** = 重力加速度
 * **acceleration — process becoming faster** = 發展 / 變化加速
 * **acceleration in growth/demand** = 增長 / 需求加快
 * **acceleration — education/progression** = 加速學習 / 進階
 * **accelerate — speed up physically** = 加速
 * **accelerate a process** = 加快 / 促進
 * **accelerate growth/development** = 加速增長 / 發展
 * **accelerated** = 加快的 / 加速的
 * **accelerated learning** = 加速學習
 * **accelerated programme** = 加速 / 密集課程
 * **accelerator — vehicle** = 油門 / 加速踏板
 * **accelerator — technical device** = 加速器
 * **particle accelerator** = 粒子加速器
 * **hardware accelerator** = 硬件加速器
 * **startup accelerator** = 創業加速器 / 加速計劃
 * **accelerator — figurative factor** = 加速因素
 * **deceleration** = 減速 / 減速能力

I would stop there. Beyond these, most further uses become specialist physics/engineering terminology or transparent compounds rather than major learner-relevant senses.

Master comparison

Form / sense	MCQ-ready Traditional Chinese meaning
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acceleration — everyday physical	人或物的移動速度變快的過程
acceleration — physics	速度向量隨時間改變的速率，包括速度大小或方向的變化
acceleration — sport	運動員在短時間內由較低速度提升至較高速度的能力
acceleration phase	由起步 / 低速逐步提升速度直至接近最高速度的運動階段
acceleration mechanics	加速期間身體姿勢、步態、發力方向及四肢協調的技術模式
linear acceleration	物體平移 / 沿路徑運動時的速度變化率
angular acceleration	物體角速度隨時間改變的速率
gravitational acceleration	物體受重力作用所產生的加速度
acceleration — process/change	某項發展、活動或變化進行速度提高的現象
acceleration in growth	增長速度本身進一步提高
acceleration — education	讓學習者比正常進度更快完成或進入更高程度
accelerate — physical	使自己 / 某物移動得更快

accelerate — figurative	令某項發展、過程或變化比原本更快發生	
accelerated	以比正常 / 先前更快速度進行的	
accelerated learning	以較正常安排更快的進度完成學習內容	
accelerator — vehicle	控制車輛增加引擎輸出並通常提高速度的油門 / 加速踏板	
accelerator — device	專門令運算、粒子或其他技術過程更快的裝置	
startup accelerator	透過資金、導師、網絡和密集支援幫助初創企業快速發展的計劃	
accelerator — figurative	令某種變革、採用或發展速度明顯提高的因素	
deceleration	移動速度降低的過程；在運動中亦指控制減速的能力	

For your original sentence

> “...a progressive programme built around mobility and **acceleration mechanics**...”

The exact meaning is:

****acceleration = 短跑運動員由低速逐步提升至較高跑速的過程 / 能力****

and:

****acceleration mechanics = 短跑運動員在加速階段的身體姿勢、落腳、步態、發力方向及四肢協調方式****

The most natural Traditional Chinese here is:

> ****「加速動作技術」****

or in a more sports-science register:

> ****「加速階段的動作力學 / 生物力學模式」****

So the passage uses the ****sporting speed-increase sense****, not the economic “growth acceleration,” educational “acceleration,” or particle-physics sense.